

The 2D Ising Model

Mathew Alex
Student Number: 7652597
PHYS 5P91

Brock University

April 9, 2024

1. 2D Ising Model

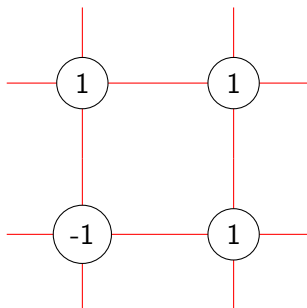
Statistical system defined by the Hamiltonian and the network topology:

$$H = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j + h \sum_i \sigma_i.$$

σ_i , often called spins or magnets, take values ± 1 .

h is analogous to the magnetic field.

J is the coupling constant.



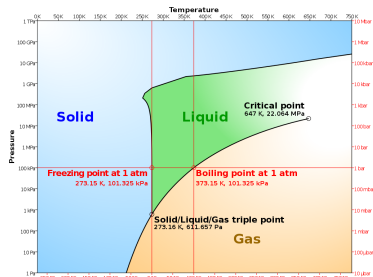
2. Why is it interesting?

1. It is real¹?
2. It has a phase transition.
3. It can make a quantum computer.

¹ "Magnetoelectric effect and critical behavior in the Ising-like antiferromagnet, DyAlO₃". In: *Solid State Communications* 9.16 (1971), pp. 1373–1376. ISSN: 0038-1098. DOI: [https://doi.org/10.1016/0038-1098\(71\)90398-X](https://doi.org/10.1016/0038-1098(71)90398-X). URL: <https://www.sciencedirect.com/science/article/pii/003810987190398X>.

2.2a What is a phase transition?

- ▶ regions on the phase diagram that can be traversed without encountering discontinuity or non-analyticity (ex: ice, water, ferromagnet, paramagnet)
- ▶ characterized by an order parameter (ex: compressibility, shear modulus, magnetization)



Phase diagram of water

[https://commons.wikimedia.org/wiki/File:](https://commons.wikimedia.org/wiki/File:Phase_diagram_of_water_simplified.svg)

[Phase_diagram_of_water_simplified.svg](https://commons.wikimedia.org/wiki/File:Phase_diagram_of_water_simplified.svg)

“I wanted to learn about elementary particles by studying boiling water.” –Alexander Markovich Polyakov

2.2c It has a PHASE TRANSITION²

1. sulfur hexafluoride liquid-gas coexistence curve,
 $|\rho_+ - \rho_-| \propto |T - T_c|^{0.327 \pm 0.006}$
2. DyAlO_3 , $M \propto (T_c - T)^{0.311 \pm 0.005}$
3. ^4He superfluid transition, $C = |T_\lambda - T|^{0.013 \pm 0.003}$
4. Biology, Economics

²N. Goldenfeld. *Lectures On Phase Transitions And The Renormalization Group*. 1992. ISBN: 9780201554090.

2.3a Quantum computers

Two types:

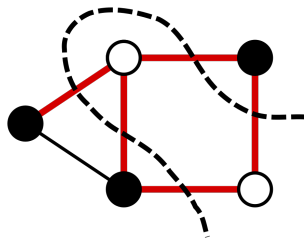
1. Turing complete computers that can run Shor's algorithm or any program in principle.
2. Annealing computer for optimization problems (ex: traveling salesman problem).

An annealing quantum computer can be made based on the Ising model³.

³Gautham Umasankar et al. "Benchmarking the Poor Man's Ising Machine". In: *2021 IEEE Photonics Society Summer Topicals Meeting Series (SUM)*. IEEE. 2021, pp. 1–2.

2.3b The max cut problem⁴

1. Problem: partition the graph into two disjoint sets such that the sum of inter-partition edge weights is maximized.
2. $H = \sum_{\text{edges}} -J_{ij}\sigma_i\sigma_j$
3. Create an Ising model like the graph with $J_{ij} = -w_{ij}$ (w_{ij} = edge weights). State minimizing the Hamiltonian is the answer.



Max-cut example

⁴Tianshi Wang et al. "Solving combinatorial optimisation problems using oscillator based Ising machines". In: *Natural Computing* 20.2 (2021), pp. 287–306.

3. All the statistical mechanics we need

The fundamental postulate of statistical mechanics:

At equilibrium, all microstates are equally likely.

$\Rightarrow$

At temperature T (or inverse temperature $\beta = \frac{1}{k_B T}$),

$$Pr(\text{state}) = \frac{e^{-\beta H(\text{state})}}{Z},$$

$$Z = \sum_{\text{all states}} e^{-\beta H(\text{state})}.$$

Z gives us everything, all the observable quantities are functions or derivatives of Z .

3.1 Also

$$\text{Energy, } \langle E \rangle = J \sum_{\langle ij \rangle} \sigma_i \sigma_j = -\frac{\partial}{\partial \beta} \ln Z,$$

$$\text{Specific heat capacity, } C_v = \frac{\partial}{\partial T} \langle E \rangle = \frac{\Delta E^2}{k_B T^2},$$

$$\text{Magnetisation, } \langle M \rangle = \sum_i \sigma_i = \frac{\partial}{\partial h} E, \text{ and,}$$

$$\text{Magnetic susceptibility, } \langle \chi \rangle = \frac{\partial}{\partial h} M = \frac{\Delta M^2}{T}$$

We will use free energy $F(T, V, N) = E(S, V, N) - TS$ (Legendre transform) $= -k_B T \ln Z$.

4. Analytic solution by Onsager

The equation⁵⁶ we are not going to use,

$$\frac{\beta F}{N} = -\ln 2 - \frac{1}{2} \int_0^{2\pi} \int_0^{2\pi} \frac{d\theta_1 d\theta_2}{(2\pi)^2} \ln[\cosh^2(2\beta J) - \sinh(2\beta J)(\cos(\theta_1) + \cos(\theta_2))].$$

Transition temperature, $T_c = \frac{2}{\ln(1+\sqrt{2})} \approx 2.27$, with $k_B = J = 1$.

⁵Lars Onsager. "Crystal statistics. I. A two-dimensional model with an order-disorder transition". In: *Physical Review* 65.3-4 (1944), p. 117.

⁶R.P. Feynman. *Statistical Mechanics: A Set Of Lectures*. Frontiers in Physics. CRC Press, 2018. ISBN: 9780429961588. URL: <https://books.google.ca/books?id=7HhQDwAAQBAJ>.

5. Monte-Carlo simulation

Idea: Drive a random spin configuration to a likely spin configuration

Algorithm

1. Initialize the spins randomly.
2. Choose a spin at random and flip it.
3. Accept the flip with a probability $e^{\beta \Delta E}$ if ΔE is negative otherwise discard.
4. Back to 2).

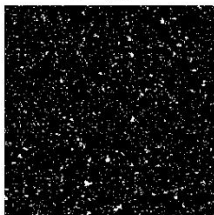
Satisfies

1. Ergodicity: The algorithm can reach all possible configurations, at least in theory
2. Detailed balance.

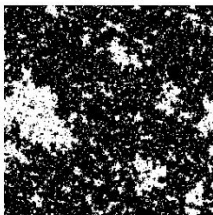
$$\implies \Pr(i) \propto e^{-\beta E_i}$$

5.1 Some Monte-Carlo snapshots for a 256×256 lattice

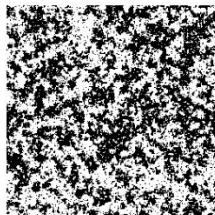
$T = 2.04$



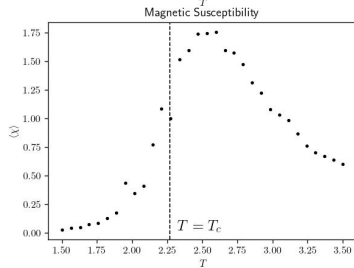
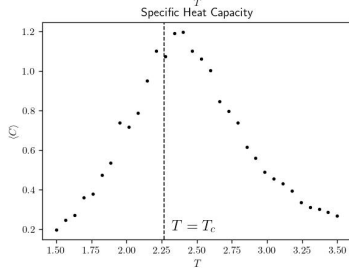
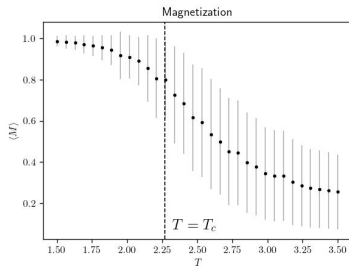
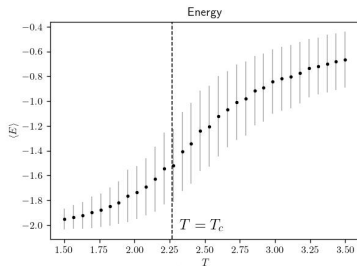
$T = 2.27$



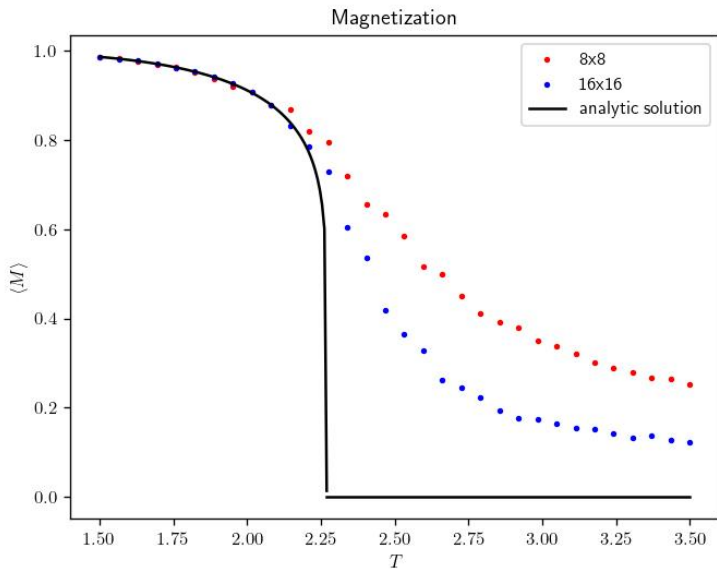
$T = 2.5$



5.2 Monte-Carlo results for an 8×8 lattice



5.3 Comparing magnetization to the analytic result



6. Tensor Networks: fancy name for contraction

In this language⁷⁸,

- ▶ a number/scalar is $\bigcirc$
- ▶ a matrix is $\text{---}\bigcirc\text{---}$, number of free edges = dimension of the tensor
- ▶ Matrix multiplication $C_{ij} = [AB]_{ij} = \sum_k A_{ik} B_{jk}$ is
 $\text{---}\bigcirc\text{---} = \text{---}\bigcirc\text{---}\bigcirc\text{---}$

Similarly, we can represent higher-order tensors and their contractions graphically.

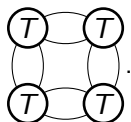
⁷Google LLC. *TensorNetwork*.

<https://github.com/google/TensorNetwork>. Accessed: 2024-04-07.

⁸Adam Jermyn. “Automatic contraction of unstructured tensor networks”. In: *SciPost Physics* 8.1 (2020), p. 005.

6.1 Partition function from Tensor Network: a happy coincidence

The partition function for 2×2 Ising model, $Z_{2 \times 2} =$



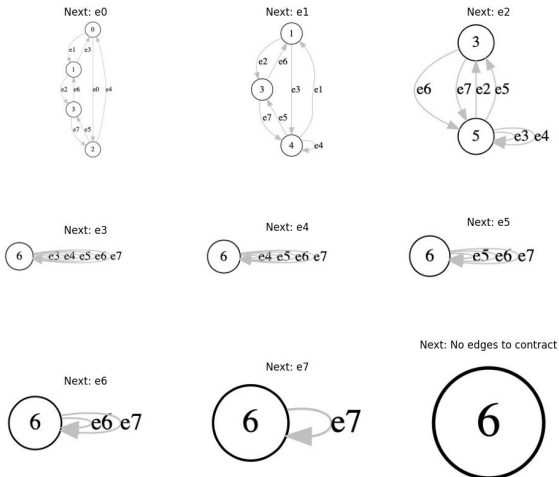
T is a 4-tensor, .

$$T_{abcd} = \sum_i S_{ia}^{\frac{1}{2}} S_{ib}^{\frac{1}{2}} S_{ic}^{\frac{1}{2}} S_{id}^{\frac{1}{2}}, \text{ where } S = \begin{bmatrix} e^{\beta} & e^{-\beta} \\ e^{-\beta} & e^{\beta} \end{bmatrix}$$

Contract all the edges to get the number $Z_{2 \times 2}$.

6.2 An example contraction

Contractions



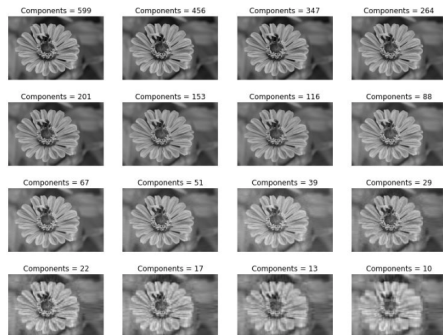
6.3 Better than counting because we can make use of linear algebra tricks

For example, Singular Value Decomposition (SVD)⁹.

Matrix, $M = U \underbrace{\Sigma}_{\text{diagonal}} V^*$

$$\text{---} \textcircled{M} \text{---} = \text{---} \textcircled{U} \textcircled{\Sigma} \textcircled{V^*} \text{---}$$

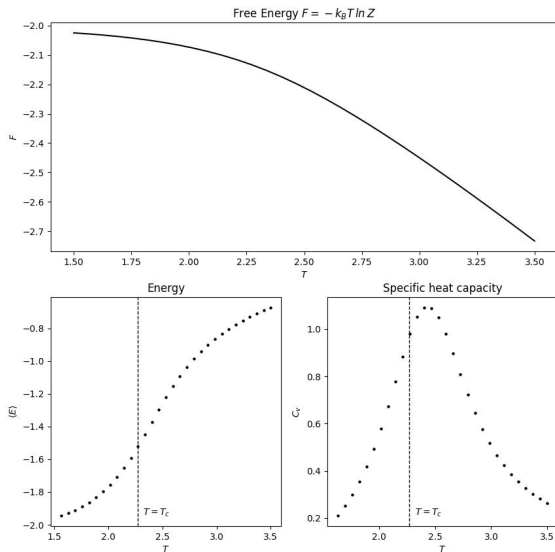
We can set the smallest few singular values (entries of Σ) to zero to reduce the computational cost.



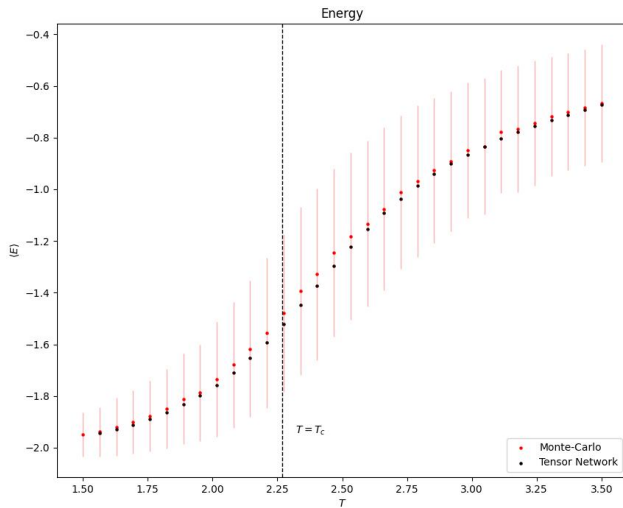
Progressive SVD approximations on an image

⁹Hai-Jun Liao et al. "Differentiable Programming Tensor Networks". In: *Physical Review X* 9.3 (Sept. 2019). ISSN: 2160-3308. DOI: 10.1103/physrevx.9.031041.

6.4 Results from the contraction ($Z_{8 \times 8}$)



6.5 A quick comparison with the Monte-Carlo result (8×8)



7. Renormalization

“The renormalization group gets little coverage in popular science articles, yet is arguably the single most important advance in theoretical physics in the past 50 years.”¹⁰– Prof. David Tong

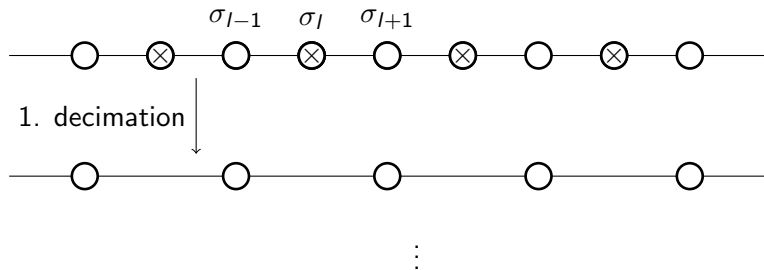
For this presentation, renormalization means,

iteratively partially evaluating Z by summing up spins, a few at a time, and trying to describe the result as an Ising model with a different coupling constant.

¹⁰David Tong. *Lectures on Statistical Field Theory*.

<https://www.damtp.cam.ac.uk/user/tong/sft.html>. Accessed: 2024-04-07.

7.1 1D Ising model, a dream come true for renormalization



Remember, $Z = \sum_{\text{all states}} e^{K \sum_I \sigma_I \sigma_{I+1}}$, where $K = \frac{J}{k_B T}$.

Contribution to Z from σ_l ,

$$\sum_{\sigma_l = \pm 1} e^{K \sigma_l (\sigma_{l-1} + \sigma_{l+1})} = 2 \cosh(K(\sigma_{l-1} + \sigma_{l+1})) =: \underbrace{e^{2g + K' \sigma_{l-1} \sigma_{l+1}}}_{\substack{\text{2. renormalisation} \\ \text{transformation}}}$$

Now find the new coupling constant K' and g .

7.1a Recursion relations

First transformation:

$$K' = \frac{1}{2} \log \cosh 2K, g = \frac{1}{2} (\log 2 + K')$$

k th transformation¹¹¹²:

$$K^{(k)} = \frac{1}{2} \log \cosh 2K^{(k-1)}, g(K^{(k)}) = \frac{1}{2} (\log 2 + K^{(k)})$$

We can do this forever on the infinite lattice. Do these values, most importantly $K^{(k)}$, approach a limit or fixed point?

Fixed points are invariant under the transformations.

¹¹F. Schwabl and W.D. Brewer. *Statistical Mechanics. Advanced Texts in Physics*. Springer Berlin Heidelberg, 2006. ISBN: 9783540323433.

¹²Humphrey J Maris and Leo P Kadanoff. "Teaching the renormalization group". In: *American journal of physics* 46.6 (1978), pp. 652–657.

7.1b Fixed points

$(K^*) = \frac{1}{2}(\log 2 + K^*)$ has two solutions:

$$K^* = 0(T = \infty) \text{ and } K^* = \infty(T = 0).$$

- ▶ Non-interacting system and fully ordered state.
- ▶ Trivial fixed points.

7.1c Partition function

$$Z_N(K) = e^{Ng(K')} Z_{N/2}(K') = e^{Ng(K') + \frac{N}{2}g(K'')} * Z_{N/2^2}(K'')$$

After n steps:

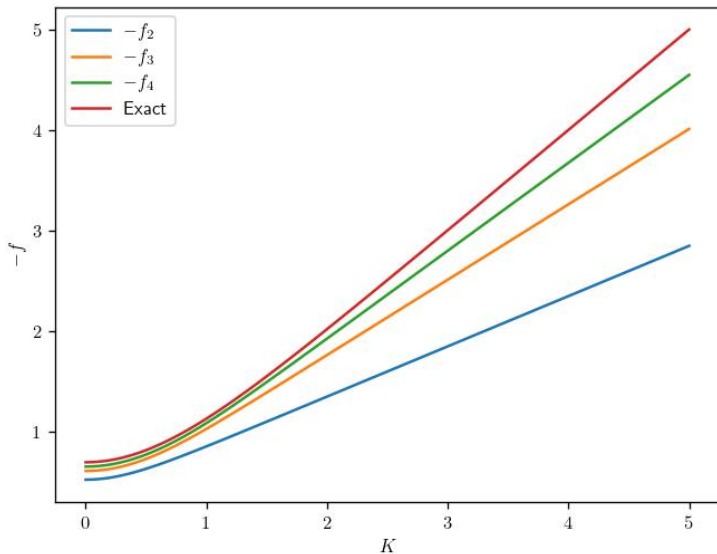
$$Z_N(K) = \exp \left[N \sum_{k=1}^n \frac{1}{2^{k-1}} g(K^{(k)}) + \ln Z_{N/2^n}(K^{(n)}) \right]$$

Free energy per lattice site and $k_B T$, $\tilde{f} = -\frac{1}{N} \log Z_N(K)$.

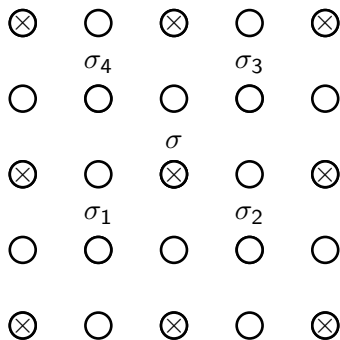
For all reasonable values of K , $K^{(n)}$ goes to zero for large n . With this,

$$\tilde{f}^{(n)}(K) = \sum_{k=1}^n \frac{1}{2^{k-1}} g(K^{(k)}) - \frac{1}{2^n} \underbrace{\log 2}_{\tilde{f} \text{ for } J=0 \text{ model}}$$

7.1d Free energy



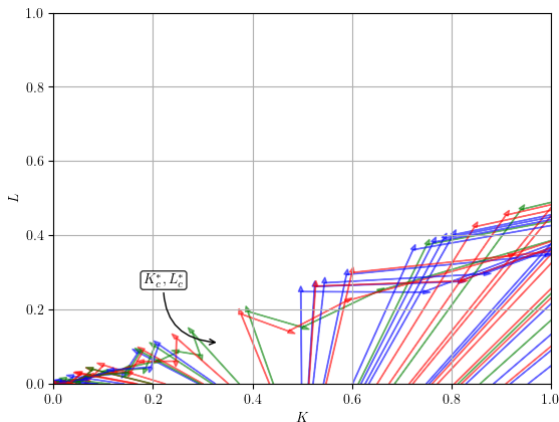
7.2 2D Ising model: A more interesting case



$$\sum_{\sigma=\pm 1} e^{K(\sigma_1+\sigma_2+\sigma_3+\sigma_4)} = 2\cosh K(\sigma_1 + \sigma_2 + \sigma_3 + \sigma_4) =$$

an expression with four constants

7.2a Flow diagram



- ▶ Non-trivial fixed points
- ▶ The 2D Ising model has a phase transition.

8. Conclusions

1. Three ways of seeing solutions for the 2D Ising Model
Monte-Carlo gives samples, noisy, no reference to Z .
Tensor Network gives Z , computationally expensive, no samples.
Renormalization: interesting, generalizable but sketchy.
2. Renormalization can be taught at undergrad level¹³?
3. “Nature uses only the longest threads to weave her patterns, so each small piece of her fabric reveals the organization of the entire tapestry.”—Richard P. Feynman

¹³Humphrey J Maris and Leo P Kadanoff. “Teaching the renormalization group”. In: *American journal of physics* 46.6 (1978), pp. 652–657.

Thank you!